

Question 1

```
Tilix: emp@cis106: ~  
1: emp@cis106: ~  
emp@cis106:~$ sudo apt install blockout2+ cuyo+ blockattack-  
The following packages were automatically installed and are no longer required:  
  libfmt10 libopusfile0 libphysfs1 libSDL2-image-2.0-0 libSDL2-mixer-2.0-0 libSDL2-ttf-2.0-0 libxmp4  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  blockout2 cuyo  
  
Installing dependencies:  
  cuyo-data libmikmod3 libSDL-mixer1.2 libSDL1.2debian  
  
REMOVING:  
  blockattack  
  
Summary:  
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 13  
  Download size: 8,078 kB  
  Space needed: 4,102 kB / 36.8 GB available  
  
Continue? [Y/n] y  
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]  
Get:2 http://deb.debian.org/debian trixie/main amd64 libSDL1.2debian amd64 1.2.68-3 [85.9 kB]  
Get:3 http://deb.debian.org/debian trixie/main amd64 libSDL-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]  
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]  
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]  
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]  
Fetched 8,078 kB in 1s (8,677 kB/s)  
(Reading database ... 195742 files and directories currently installed.)  
Removing blockattack (2.9.0-1+b1) ...  
Selecting previously unselected package libmikmod3:amd64.  
(Reading database ... 195715 files and directories currently installed.)  
Preparing to unpack .../0-libmikmod3-3.3.13-1_amd64.deb ...  
Unpacking libmikmod3:amd64 (3.3.13-1) ...  
Selecting previously unselected package libSDL1.2debian:amd64.  
Preparing to unpack .../1-libSDL1.2debian-1.2.68-3_amd64.deb ...  
Unpacking libSDL1.2debian:amd64 (1.2.68-3) ...  
Selecting previously unselected package libSDL-mixer1.2:amd64.  
Preparing to unpack .../2-libSDL-mixer1.2-1.2.12-19+b1_amd64.deb ...  
Unpacking libSDL-mixer1.2:amd64 (1.2.12-19+b1) ...  
Selecting previously unselected package blockout2.  
Preparing to unpack .../3-blockout2-2.5+dfsg1-2_amd64.deb ...  
Unpacking blockout2 (2.5+dfsg1-2) ...  
Selecting previously unselected package cuyo-data.  
Preparing to unpack .../4-cuyo-data-2.1.0-1_all.deb ...  
Unpacking cuyo-data (2.1.0-1) ...  
Selecting previously unselected package cuyo.  
Preparing to unpack .../5-cuyo-2.1.0-1_amd64.deb ...  
Unpacking cuyo (2.1.0-1) ...  
Setting up libmikmod3:amd64 (3.3.13-1) ...  
Setting up libSDL1.2debian:amd64 (1.2.68-3) ...  
Setting up libSDL-mixer1.2:amd64 (1.2.12-19+b1) ...  
Setting up blockout2 (2.5+dfsg1-2) ...  
Setting up cuyo-data (2.1.0-1) ...  
Setting up cuyo (2.1.0-1) ...  
Processing triggers for libc-bin (2.40-3) ...
```

```
1/1 v + G T Tilix: emp@cis106: ~
emp@cis106:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
cuyo-data libfmt10 libopusfile0 libphysfs1 libSDL2-image-2.0-0 libSDL2-mixer-2.0-0 libSDL2-ttf-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
cuyo

Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 13
Freed space: 489 kB

Continue? [Y/n] y
(Reading database ... 196812 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
emp@cis106:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
cuyo-data libfmt10 libopusfile0 libphysfs1 libSDL2-image-2.0-0 libSDL2-mixer-2.0-0 libSDL2-ttf-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
cuyo

Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 13
Space needed: 0 B / 36.8 GB available

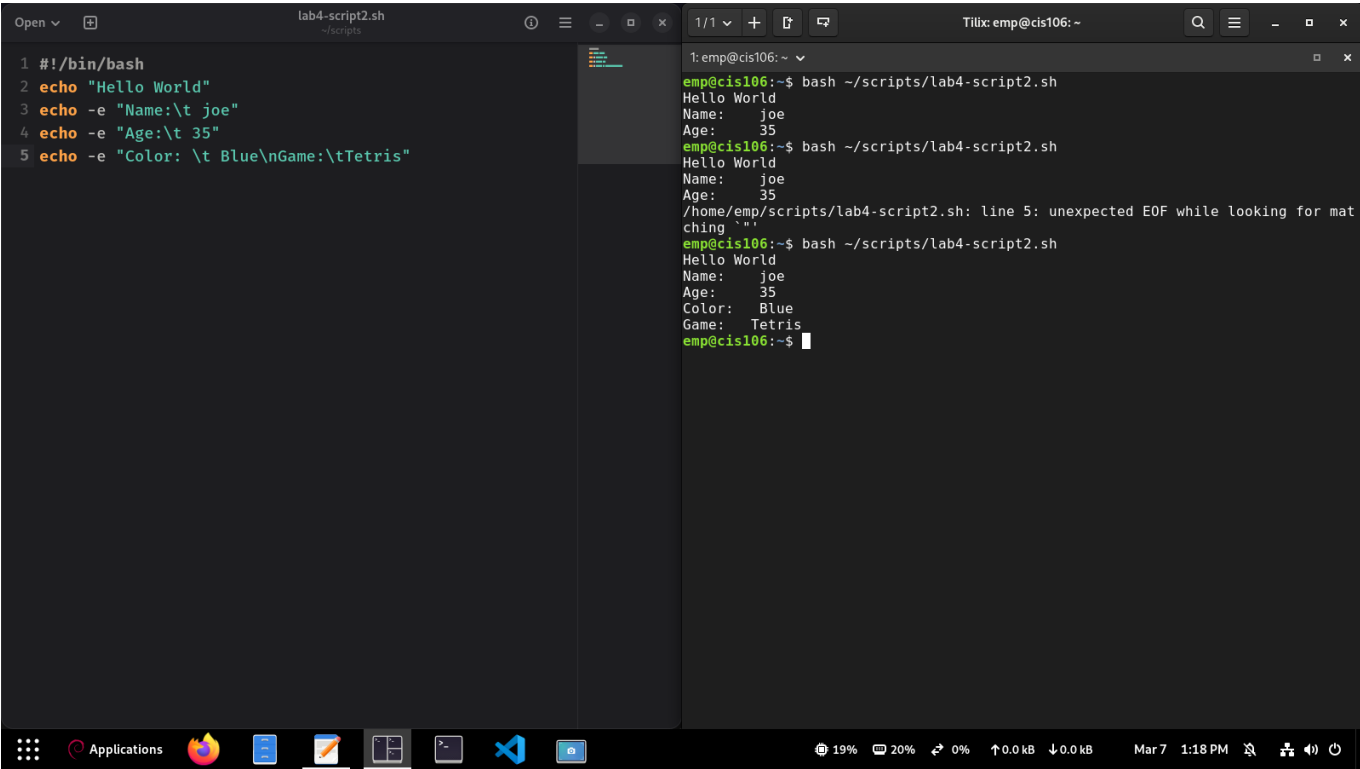
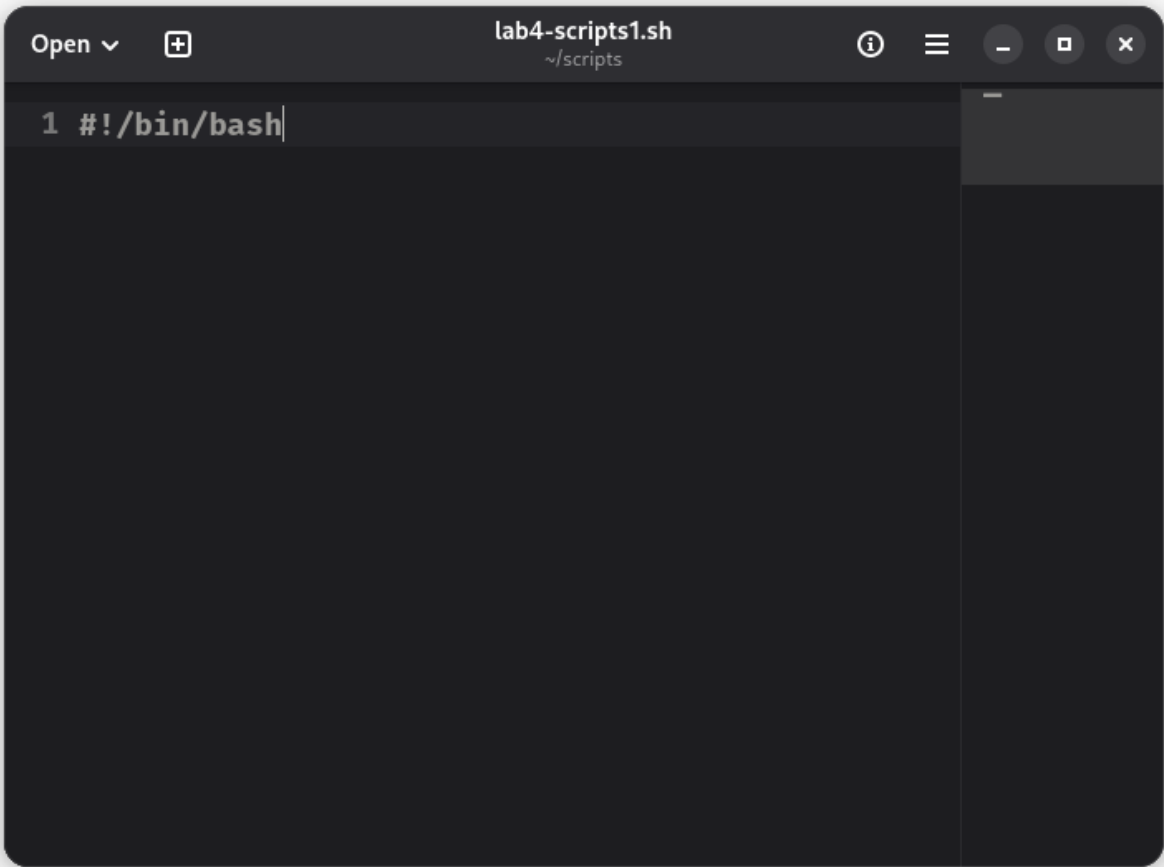
Continue? [Y/n] y
(Reading database ... 196804 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1) ...
emp@cis106:~$
```

```
1/1 v + G T Tilix: emp@cis106: ~
emp@cis106:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
cuyo-data libfmt10 libopusfile0 libphysfs1 libSDL2-image-2.0-0 libSDL2-mixer-2.0-0 libSDL2-ttf-2.0-0 libxmp4

Summary:
Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 13
Freed space: 7,495 kB

Continue? [Y/n] y
(Reading database ... 196804 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libSDL2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12.4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libSDL2-image-2.0-0:amd64 (2.8.8+dfsg-1) ...
Removing libSDL2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 13
emp@cis106:~$
```

Question 2



lab4-script3.sh
~/scripts

```
1 #!/bin/bash
2 echo "Today's date is: "
3 date
4 echo -n "The current user is "
5 whoami
6 # echo -n "The current user is "; whoami
7 echo "Memory Usage: "
8 free -h
```

1: emp@cis106: ~

```
emp@cis106:~$ bash ~/scripts/lab4-script3.sh
Today's date is:
Sat Mar 7 01:33:05 PM EST 2026
The current user is emp
Memory Usage:

```

	total	used	free	shared	buff/cache	availabl
Mem:	7.7Gi	1.5Gi	4.8Gi	13Mi	1.6Gi	6.2G
Swap:	2.6Gi	0B	2.6Gi			

```
emp@cis106:~$
```

Question 3

challenge_lab4.sh
~/scripts

```
25 }
26
27 echo ""
28 figlet "DISK" 2>/dev/null || echo "DISK"
29 echo "$LINE"
30
31 echo "Filesystem      Size  Used Avail Use% Mounted on"
32 df -h / | tail -n 1 | awk '{printf "%-15s %5s %5s
33 %5s %4s %s\n", $1, $2, $3, $4, $5, $6}'
34 echo ""
35 echo "$LINE"
```

emp@cis106: ~

```
emp@cis106:~$ bash ~/cis106/labs/lab4/challenge_lab4.sh

```

```

  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _
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```

```

DATE UTC:      03/07/26 18:47:07
HOSTNAME:      cis106
UPTIME:        3:37
USER:          emp

```

```

  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _
 /  /  /  /  /  /  /  /  /  /  /  /  /  /  /  /  /  /  /  /
/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/

```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	47G	8.9G	36G	21%	/

```
emp@cis106:~$
```

script